

100 Days of Linux - Day 057

Title: Practical Assignment - Linux Networking

Today's Goal

Apply everything learned in the Networking module using realistic Linux administration tasks. No new commands are introduced today.

Why This Matters

Networking skills are essential for developers, cloud engineers, DevOps engineers, and data engineers. You'll verify connectivity, inspect network settings, download data, and prepare secure remote access.

Commands Covered

ping, curl, wget, ssh, scp, ip

Scenario

You have been asked to verify a new Linux server before deploying a data pipeline. Check connectivity, inspect the network, download reference files, and prepare secure access.

Practical Assignment (5 Tasks)

1. Use **ip addr** and **ip route** to identify your primary interface, IP address, and default gateway. Save the information in `network_report.txt`.
2. Ping localhost, 8.8.8.8, and google.com using **-c 4**. Compare the average response times and note any packet loss.
3. Use **curl** to download `https://example.com` into `example.html`. Then use **wget** to download the same page into a folder named `downloads` using a custom filename.
4. Check whether you already have SSH keys by listing `~/.ssh`. Write the SSH command you would use to connect to a server at `192.168.1.100` with username `ubuntu`, then write the SCP command to upload `report.txt` to that server.
5. Display your current working directory and list the files you created during today's assignment.

Hints

Complete one task at a time and verify the output after each command. Today's goal is to combine multiple networking commands into one workflow.

Mini Challenge

Create a folder named `deployment_checklist` containing `network_report.txt`, `example.html`, and a file named `ssh_commands.txt` with sample SSH and SCP commands. Compress the folder into `deployment_checklist.zip`.

Common Mistakes

Using `curl` when you only need a file download, or forgetting that

SSH and SCP require a reachable remote server to establish a connection.

Did You Know?

Many CI/CD pipelines use curl to call APIs, wget to fetch artifacts, SSH to access servers, and SCP to deploy application files.

Pro Tip

When troubleshooting networking, start locally with ip and ping before attempting remote connections with SSH or SCP.

Answer Key (Instructor Only)

Q1: ip addr && ip route

Q2: ping -c 4 localhost && ping -c 4 8.8.8.8 && ping -c 4 google.com

Q3: curl -o example.html https://example.com && mkdir -p downloads && wget -O downloads/example.html https://example.com

Q4: ls -la ~/.ssh && ssh ubuntu@192.168.1.100 && scp report.txt ubuntu@192.168.1.100:/home/ubuntu/

Q5: pwd && ls

Homework

Repeat today's workflow from memory until you can comfortably use each networking command without referring to notes.

Next Day Preview

Tomorrow you'll begin the Environment Variables module by learning how variables work in the Linux shell using printenv, env, and echo.